MARINE C. CAMBON

Research Fellow in Microbial Ecology

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POSITIONS

Research Fellow and UK Treescapes Fellow (current)

The University of Birmingham

April 2023 – December 2025 | Birmingham, UK

Data-driven design of a synthetic community of the oak microbiota to tackle Acute Oak Decline

- Design and in vitro assembly of a healthy oak microbiota SynCom
- Investigation of the ecological processes at play when an AOD pathogen invades the host microbiota community

Postdoctoral researcher

Bangor University

December 2020 – March 2023 | Bangor, Gwynedd, UK

Understanding multi-kingdom interactions in Acute Oak Decline
BAC-STOP project website

- Can we identify the chemical cues triggering virulence gene expression in AOD?

Postdoctoral researcher

Lab. Biodiversité Gènes Communautés (UMR INRAE-UB)

May 2019 – December 2020 | Bordeaux, France

Next-generation biomonitoring of changes in ecosystems
ANR project NGB website

- Can we predict tree water stress from leaf microbiota using machine learning techniques?

Side project: Drought resistance and leaf microbial composition in tropical forest trees
DROUGHT project, Labex CEBA

EDUCATION

PhD in microbial ecology and evolution

EDB lab (CNRS-UPS) and DGIMI lab (INRA-UM)

October 2015 – December 2018 | Toulouse and Montpellier, France

Heterogeneity in bacterial infections: the case of *Xenorhabdus nematophila*, an vector-borne pathogen of insect

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MRes degree, Biology Ecology and Evolution

Paul Sabatier University

2013 – 2015 | Toulouse, France

BSc degree, Organism Population and Ecosystem Biology

Paul Sabatier University

2011 – 2013 | Toulouse, France

SCIENTIFIC INTERESTS

Microbial ecology & evolution | Microbiota | Symbiosis | SynComs | Host-pathogen interactions

SKILLS

Experimental work with insects | qPCR | Amplicon sequencing | Transcriptomics | Metagenomics | Microbiology | Field sampling | Plant microcosms

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Statistics | Bioinformatic | Unix | R | Markdown | Git | LaTeX

GRANTS

- **UK Treescapes Fellowship (£32,500)**, NERC (2023). *Developing oak seedling microcosms to develop SynComs of the oak microbiota*
- **NEOF Pilot Project Grants (£12,000)**, NERC (2023). *Genome sequencing for data-driven design of an oak microbiota SynCom*
- **ECDN Career Development Competition (£500)**, University of Birmingham (2023). *Visiting collaborators in the University Hospital RWTH Aachen, Germany*
- **School of Bioscience Pump priming fund (£5,000)**, University of Birmingham (2023). *Taxonomic characterisation of pak isolate collection for data-driven design of an oak microbiota SynCom*
- **Research support grant (£1,000)**, Bangor University (2021). *Development of as qPCR assay to quantify virulence gene expression in* Brenneria goodwinii

AWARDS

- Applied Microbiology International prize for the best talk, MMEG conference 2024
- FEMS Microbiology letters prize for the best flash presentation, MMEG conference 2021

RESEARCH INTERNSHIPS

Master research project

The role of phenotypic switching of *Xenorhabdus nematophila*

2015 — UMR EDB, Toulouse

Master internship

Modeling the link between primary productivity and food chain length in lakes

2014 — UMR EDB, Toulouse

Bachelor internship

Ecology of fish parasites community : statistical analysis of ecological and epidemiological parameters

2013 — Lab. Arago, Banyuls-sur-mer

Bachelor internship

Reproductive system variability of an alpine plant *Rhododendron ferrugineum* at the Pyrénées scale

2013 — UMR EDB, Toulouse

CONFERENCES

Most recent communications:

- **MMEG conference** – 2024, *Southampton, UK* (**Talk**).
- **ISME conference** – 2024, *Cape Town, South Africa* (**Short talk**)
- **REID conference** (Long-lasting interactions network) – 2023, *Poitiers, France* (**Talk**)
- **Microbiology Society conference** – 2023 *Birmingham* (**Talk**)
- **ISME conference** – 2022, *Lausanne, Switzerland* (**Poster**).
- **MMEG conference** – 2021, *Norwich, UK* (**Flash talk**).
- **Phyllosphere conference** – 2021, *Online* (**Talk**)
- **Plant-Bacteria biannual meeting** – 2020, *Aussois, France* (**Poster**)
- **ESEB congress** (European Society for Evolutionary Biology) – 2019, *Turku, Finland* (**Talk**)
- **Microbioccitanie conference** – 2019, *Montpellier, France* (**Talk**)
- **Microbial Eco-Evo Dynamics symposium** – 2018, *Oeiras, Portugal* (**Poster**)
- **Second Joint Congress on Evolutionary Biology** – 2018, (**Talk**)
- **IMH annual meeting** (Host-Microorganism Interactions network) – 2018, (**Talk**)

TEACHING

Grant writing and scientific communication

MRes students

2023-2024 — University of Birmingham

Machine learning and molecular data, plant pathology

2nd and 3rd year undergrad students, master students

2021 – 2022 — Bangor University

Statistics and modeling for ecology, 64h/year

2nd and 3rd year B.Sc students

2015 – 2018 — Univeristé Paul Sabatier

MENTORSHIP

Postgraduate students

- **Usman Hussain**. PhD candidate (2021-2024)

Undergraduate students

- **Ananya Singh**. MSc student (2024)
- **Inderpreet Singh**. MSc student (2024)
- **Muyuan He**. MSc student (2024)
- **Upasana Sharma**. MSc student (2024)
- **Yves Ritter**. BSc student (2019)
- **Pierre Lafont**. BSc student (2017)
- **Adrian Bach**. BSc student (2016)

TEACHING QUALIFICATION

- Associate Fellow of Higher Education Academy (AFHEA) qualification: application submitted in January 2025.

PEER-REVIEW WORK

- Microbiome
- Molecular ecology
- PeerJ
- Proceedings of the Royal Society Biology
- Agriculture, Ecosystems and Environment
- Plant Pathology

EQUALITY, DIVERSITY AND INCLUSION ACTIVITIES

- Member of the Sexual Violence Disclosure Response Team (Bangor University)
- Member of he Rainbow Network for LGBTQ+ inclusion (University of Birmingham)

CITIZENSHIP ACTIVITIES

Organisation of the BIFoR annual meeting

2024 — Birmingham, UK

Organisation of the REID conference

2017 — Toulouse, France

Organisation of the department journal club for ECRs

2015-2017 — Toulouse, France

LIST OF PUBLICATIONS

Peer-reviewed articles

- Fournier P, Pellan L, Jaswa A, **Cambon MC**, Chataigner A, Bonnard O, Raynal M, Debord C, Poeydebat C, Labarthe S, Delmotte F, This P, Vacher C (2025). Revealing microbial consortia that interfere with grapevine downy mildew through microbiome epidemiology. Environmental Microbiomes. *In press.*
- Hussain U, Downie J, Ellison E, Denman S, McDonald J, **Cambon MC** (2025) Peptide nucleic acid (PNA) clamps reduce amplification of host chloroplast and mitochondria rRNA gene sequences and increase detected diversity in 16 S rRNA gene profiling analysis of oak-associated microbiota. Environmental Microbiomes 20, 14. https://doi.org/10.1186/s40793-025-00674-w
- Ordonez A, Hussain H, **Cambon MC**, Downie J, McDonald J (2025) Agar-plating and dilution-to-extinction based microbial isolation approaches capture different fractions of the oak microbiota. ISME Communications ycaf019. https://doi.org/10.1093/ismeco/ycaf019
- **Cambon MC**, Trillat M, Lesur-Kupin I, Burlett R, Chancerel E, Guichoux E, et al. (2023) Microbial biomarkers of tree water status for next-generation biomonitoring of forest ecosystems. Molecular Ecology 32: 59445958. http://dx.doi.org/10.1111/mec.17149
- **Cambon MC**, Cartry D, Chancerel E, Ziegler C, Levionnois S, Coste S, Stahl C, Delzon S, Buée M, Burban B, Cazal J, Fort T, Goret J-Y, Heuret P, Léger P, Louisanna E, Ritter Y, Bonal D, Roy M, Schimann H, Vacher C (2022). Drought tolerance traits in Neotropical trees correlate with the composition of phyllosphere fungal communities. Phytobiome e-ISSN:2471-2906. http://dx.doi.org/10.1094/PBIOMES-04-22-0023-R
- Côte J, Jacquin L, Veyssière C, Manzi S, Etienne R, Perrault A, **Cambon MC**, Jean S, White J (2022). Changes in fish skin microbiota along gradients of eutrophication in human-altered rivers. FEMS Microbiology Ecology 98(1). http://dx.doi.org/10.1093/femsec/fiac006.
- Dubart M, Alonso P, Barroso-Bergada D, Becker N, Bethune K, Bohan DA, Boury C, **Cambon MC**, et. al, Massol F (2021). Chapter Nine - Coupling ecological network analysis with high-throughput sequencing-based surveys: Lessons from the next-generation biomonitoring project. Advances in Ecological Research 65:367-430. https://doi.org/10.1016/bs.aecr.2021.10.007
- Compant S, **Cambon MC**, Vacher C, Mitter B, Samad A, Sessitsch A (2020). The plant endosphere world – bacterial life within plants. Environmental Microbiology 23: 1812-1829. https://doi.org/10.1111/1462-2920.15240
- **Cambon MC**, Lafont P, Frayssinet M, Lanois A, Ogier J-C, Pagès S, Parthuisot N, Ferdy J-B, Gaudriault S (2020). Bacterial community profile after the lethal infection of *SteinernemaXenorhabdus* pairs into soil-reared *Tenebrio molitor* larvae. FEMS Microbiology Ecology 96:fiaa009. https://doi.org/10.1093/femsec/fiaa009
- **Cambon MC**, Parthuisot N, Pagès S, Lanois A, Givaudan A, Ferdy J-B (2019). Selection of bacterial mutants in late infections: when vector transmission trades off against growth advantage in stationary phase. mBio 10:e01437-19. https://doi.org/10.1128/mBio.01437-19
- Payelleville A, Blackburn D, Lanois A, Pagès S, **Cambon MC**, Ginibre N, Clarke DJ, Givaudan A, Brillard J (2019). Role of the *Photorhabdus* Dam methyltransferase during interactions with its invertebrate hosts. PLOS ONE 14(10):e0212655. https://doi.org/10.1371/journal.pone.0212655

Pre-prints and submitted manuscripts

- **Cambon MC**, Thomas G, Crampton M, Caulfield JC, Doonan J, Hussain U, Denman S, Vuts J, McDonald J. Chemical cues trigger *Brenneria goodwinii* pathogenicity in Acute Oak Decline. *Under review in Proc. B.* bioRxiv https://doi.org/10.1101/2023.11.21.568124
- Hussain U, **Cambon MC**, Ordoñez A, Downie J, Brown N, Finch J, Beckmann M, Vanguelova E, Elison A, Brady C, Denman S, McDonald JE (2025). Resilience of semi-mature oak microbiota to biotic and abiotic perturbations. *Submitted to Nature Communications*
- Downie J, Ordonez A, Hussain U, **Cambon MC**, Brown N, Denman S, McDonald J. Future Oak: a microbiome-wide association study of acute oak decline severity in Quercus robur